1 Prime Numbers and Unique Factorization

1.1 Primes

Definition 1.1.1

An integer $p \in (\mathbb{Z} \setminus \{0, \pm 1\}$ is said to be **prime** if its only divisors are 1, -1, p, and $-p$. A **prime number** is a natural number which is prime, i.e. a number $p \geq 2$ such that its only divisors are 1 and p.

Note that by definition, 1 is not a prime number.

Prime numbers have very important and unique properties and are thus of great interest to the math world. Indeed some of the most famous unsolved problems, such as the twin primes conjecture, are centred around prime numbers. Finding prime numbers is also of great interest; indeed finding the largest known prime number can yield one a (fairly) hefty sum of cash.

1.2 The Sieve of Eratosthenes

To find prime numbers, we can use the *Sieve of Eratosthenes*, which is a slow and literally ancient, yet reliable algorithm to find all prime numbers from 2 to a natural number N.

Suppose, for sake of exposition, we want to find all the prime numbers between 2 and $N = 50$. We first write out all numbers between 2 and 50:

2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28
29	30	31	32	33	34	35	36	37
38	39	40	41	42	43	44	45	46
47	48	49	50					

We start by highlighting the smallest number–in this case it is 2. 2 is a prime number. We then cross out all multiples of 2, as they are not prime:

2	3	~~4~~	5	~~6~~	7	~~8~~	~~9~~	~~10~~
11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19
~~20~~	~~21~~	~~22~~	23	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~
29	~~30~~	31	~~32~~	~~33~~	~~34~~	~~35~~	~~36~~	37
~~38~~	~~39~~	~~40~~	41	~~42~~	43	~~44~~	~~45~~	~~46~~
47	~~48~~	~~49~~	~~50~~					

Now, we move to highlight the next smallest number: 3. We now cross out all multiples of 3 (we

can skip numbers that have already been crossed out):

2	3	~~4~~	5	~~6~~	7	~~8~~	~~9~~	~~10~~
11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19
~~20~~	~~21~~	~~22~~	23	~~24~~	25	~~26~~	~~27~~	~~28~~
29	~~30~~	31	~~32~~	~~33~~	~~34~~	35	~~36~~	37
~~38~~	~~39~~	~~40~~	41	~~42~~	43	~~44~~	~~45~~	~~46~~
47	~~48~~	~~49~~	~~50~~					

The next smallest number, 4, has been crossed out (hence is not prime), so we move to the next lowest number, 5, which is prime. We now cross out all multiples of 5:

2	3	~~4~~	5	~~6~~	7	~~8~~	~~9~~	~~10~~
11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19
~~20~~	~~21~~	~~22~~	23	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~
29	~~30~~	31	~~32~~	~~33~~	~~34~~	~~35~~	~~36~~	37
~~38~~	~~39~~	~~40~~	41	~~42~~	43	~~44~~	~~45~~	~~46~~
47	~~48~~	~~49~~	~~50~~					

Continuing this algorithm until all numbers have been either crossed out or highlighed, we get the prime numbers up to 50:

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47

1.3 Properties of Prime Numbers

Theorem 1.3.2 (Characterisation of Primes)

Let $p \geq 2$ be a natural number. p is a prime number if and only if for all $a, b \in \mathbb{Z}$, if $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. Assume that p is prime and that for all $a, b \in \mathbb{Z}$, $p \mid ab$. Assume $p \nmid a$, then $\gcd(a, p) = 1$ since p is prime. Thus, by Corollary 6.1.3, $p \mid b$. The same logic works analogously if we assume $p \nmid b$. Conversely, suppose that for all $a, b \in \mathbb{Z}$, $p \mid ab$ implies $p \mid a$ and $p \mid b$. We suppose $p = st$ for some $s, t \in \mathbb{N}^+$ and show that $s = 1$ or $t = 1$. Since $p \mid st$, $p \mid s$, or $p \mid t$, say $p \mid s$. Then $s = ps'$ hence $p = st = s't$ so $s't = 1$, hence $s' = t = 1$. ∎

Corollary 1.3.3

If $p \geq 2$ is prime and $p \mid a_1 a_2 \dots a_n$ for $a_1, \dots, a_n \in \mathbb{Z}$, then there exists an $i \leq n$ such that $p \mid a_i$.

Proof. We proceed by induction on n. If $n = 1$, then the statement $p \mid a_1$ holds with $i = 1$. Now suppose the statement holds for n integers, i.e. $p \mid a_1 a_2 \dots a_n$. Suppose $p \mid a_1 a_2 \dots a_n a_{n+1}$. We can write this as:

$$p \mid (a_1 a_2 \dots a_n) a_{n+1}$$

Since p is prime, we have by Theorem 7.3.2 that $p \mid (a_1 a_2 \dots a_n)$ or $p \mid a_{n+1}$. If $p \mid a_{n+1}$ we are done. Otherwise, by the induction hypothesis, we have that there exists an $i \leq n$ such that $p \mid a_i$. Thus the statement holds by induction. ■

Theorem 1.3.4

Every natural number $n \geq 2$ is a product of primes.

Proof. We proceed by complete induction on n. The base case $n = 2$ is trivial as 2 is a prime number. For the inductive step, let $k \geq 2$ and assume that every $2 \leq n \leq k$ can be written as a finite product of prime numbers. We want to show that $k + 1$ can be written as a product of primes. Without loss of generality assume $k + 1$ is composite. By definition of divisibility, $k + 1$ can be written as the product of two natural numbers a and b where $a, b \geq 2$. By the induction hypothesis, we can write both a and b as a finite product of prime numbers, hence $k + 1$ can be written as the product of finitely many prime numbers. ■

Corollary 1.3.5

Every natural number $n \geq 2$ is divisible by a prime.

Proof. Analogous to the proof of Theorem 7.3.4. ■

Corollary 1.3.6

There are infinitely many prime numbers.

Proof. Suppose by contradiction that there are finitely many primes $p_1, \dots, p_k$ for some $k \geq 1$. Let $n := p_1 p_2 \dots p_k + 1$. By Corollary 7.3.5, there exists a prime number q such that $q \mid n$, hence so $q = p_i$ for some $i \in \{1, \dots, k\}$ since $p_1, \dots, p_k$ are all the primes. But then $1 = n - p_1 \dots p_k$ is divisible by $q = p_i$ since $q \mid a$ and $q \mid p_1 \dots p_k$, which is a contradiction. ■

Theorem 1.3.7 (Fundamental Theorem of Arithmetic)

Every nonzero integer n is a unique product of prime integers, i.e. $n = \sigma p_1 \dots p_k$, where $\sigma = \pm 1$ and $2 \leq p_1 \leq \dots \leq p_k$ are primes.

Proof. The existence of such a representation follows from Theorem 7.3.4. We only need to show that this representation is unique.

Let $\sigma p_1 \dots p_k = n = \sigma' q_1 \dots q_m$ where p_i and p_j are prime numbers ordered $2 \leq p_1 \leq \dots \leq p_k$ and $2 \leq q_1 \leq \dots \leq q_m$. The sign must be the same, i.e. $\sigma = \sigma'$, so assume without loss of generality that $n \geq 0$ and $p_1 \dots p_k = n = q_1 \dots q_m$. We prove that $k = m$ and $p_i = q_i$ for all $1 \leq i \leq k$. We can show this holds by strong induction on $n \geq 1$.

If $n = 1$, then $k = 0 = m$ and we are done. If we suppose $n \geq 2$, then $k, m \geq 1$ and without loss of generality suppose $p_1 \leq q_1$. Then $p_1 \mid q_1 \dots q_m$, so $p_i \mid q_j$ for some $1 \leq j \leq m$. But q_j is prime so

$p_1 = q_j$. Since $q_1 \leq \ldots q_j$ and $p_1 \leq q_1$, we get $p_1 = q_1 = \cdots = q_j$, so $n' := p_2 p_3 \ldots p_k = q_2 q_3 \ldots q_m$. By the induction hypothesis on n', we get $k = m$ and $p_i = q_i$ for all $i \in [2, k]$, hence $k = m$ and $p_i = q_i$ for all $i \in [2, k]$, whence the representation is unique. ■